REVEAL: Agentic In-Hand Manipulation for Language-Guided Target Presentation

Anonymous CVPR submission

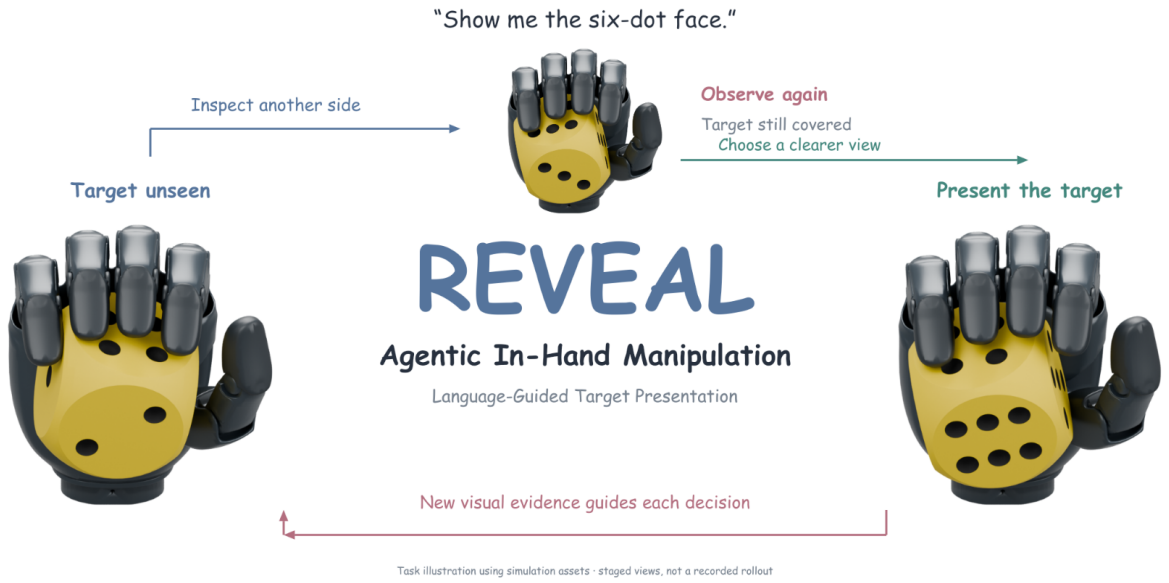


Figure 1. **REVEAL overview.** New visual evidence guides semantic decisions as the hand inspects and presents a requested object face. This staged simulation illustration is not a recorded rollout or physical-robot result.

Abstract

How can a robot find and present a requested object face when it is initially hidden or blocked by the fingers? We present REVEAL, an agentic in-hand manipulation framework connecting visual search, target binding and continuous control. The agent chooses inspection actions, uses new observations to identify the target, and selects a motor goal. A ViserDex-inspired teacher–student system adapted to Revo3 executes these goals, with local controllers adjusting joint commands during reasoning and execution. Paired cube experiments show that fresh images increase success from 9/40 to 32/40 by enabling additional correct bindings. A three-seed controller study supports joint correction during inference. Across six object-specific simulation systems, changed-mapping success averages 65.8%, exceeding the strongest evaluated baseline by 40.4 percentage points. English and Chinese instructions are evaluated on three objects.

1. Introduction

A command such as "present the face marked T" tells the robot what to show. The robot still has to find T, choose a motor goal that exposes it, and move the object without dropping it. The target may initially face away from the camera or be blocked by the fingers. Moving the object can reveal the target, but also changes the grasp. The robot therefore needs to decide both what to look at next and how to keep controlling the object.

Vision-language agents can interpret the instruction and inspect camera images, but each response takes time. Meanwhile, the hand and object continue to move. Repeating the previous command may lose the grasp or leave the target hidden. Slowing down limits motion but cannot change the direction of the joint command. A local controller must therefore respond to new measurements while the agent is still reasoning.

REVEAL lets the agent and motor policy run at different rates while sharing the same motor goals. The agent chooses inspection actions from the current image and action history.

Once it identifies the requested face, it links that face to a motor goal, a step we call *target binding*. A recurrent student uses this goal, measured pose and proprioception to propose joint commands. For the cube, two local controllers adjust these commands during agent inference and action execution. Both learn from teacher actions in simulation. The hand can thus respond to its current state before the next agent reply arrives.

Our motor foundation follows the teacher–student approach of ViserDex [4], adapted to the Revo3 dexterous hand. REVEAL builds on this foundation with a shared goal bank, explicit inspection and target-binding decisions, and local control during agent inference. A fine-tuned Qwen3-VL [2] predicts a goal vector directly, while a rule-based VLM returns a color and a readability decision. Within each object, these systems share the motor student but differ in inspection, history, local control and stopping. We compare their complete-system performance and use controlled cube studies to examine individual components.

Paired cube studies examine how the agent acquires target evidence. Fresh images enable 25 additional correct bindings after the first call, with 23 leading to success; repeating the first image enables none. A new comparison tests adaptive against fixed inspection while keeping the model and controllers unchanged. Its 17.5-point observed success gain remains statistically unresolved. A separate three-seed study supports joint correction during agent inference.

Across 1,440 simulation trials on six objects, REVEAL achieves 70.4% success with the original target-to-skill mappings and 65.8% after the mappings change. These equal-object averages exceed the strongest baseline by 26.2 and 40.4 percentage points, respectively. Each object has its own policy and perception setup. A separate 2,160-trial English/Chinese study favors REVEAL over fine-tuning in all eight duck and hedgehog conditions after correction for multiple comparisons; cube differences remain inconclusive.

Our contributions are threefold. First, we study language-guided target presentation through visual search, target binding and continuous in-hand control. Second, we connect an agent to a recurrent motor policy through explicit goal selection and teacher-supervised local controllers. Third, we evaluate six object-specific systems and two languages, with cube studies testing continued feedback, fresh images, inspection choices and waiting-time control.

2. Related Work

RL for dexterous in-hand manipulation. RL policies trained in simulation can reorient objects with a physical hand using domain randomization [1] and handle diverse object shapes [7]. Rapid motor adaptation handles changing object properties [15]. RotateIt combines vision, touch and proprioception through teacher–student distillation [16], while AnyRotate supports different rotation axes and hand orientations [21]. REVEAL builds on these motor capabilities to find and present a target specified by language, even when the robot must first move the object to see it.

Visual and goal-conditioned dexterity. ViserDex combines monocular pose estimation with recurrent teacher–student control [4]. Our Revo3 motor foundation adapts this approach; REVEAL adds goal-bank conditioning, agent-directed search and local control during reasoning. GoalDex uses current and goal images, selects goals from language with a VLM, and studies multi-object sim-to-real transfer and visual-processing delays [3]. Our focus is acquiring the evidence needed to identify a requested face while hand control continues. ManipTrans uses residual learning for dexterous bimanual transfer [12]; our local corrections support inspection and presentation within one hand.

Agentic reasoning and embodied feedback. SayCan selects language-guided skills using robotic affordances [11], and Inner Monologue uses environmental feedback to update plans [9]. Code as Policies connects perception and control APIs [14]. VoxPoser and ReKep translate language into spatial value maps and keypoint constraints [8, 10]. Magma connects multimodal representations to actions [20], while VERA includes language-guided dexterous reorientation from visual plans [13]. REVEAL studies this interaction for a target held in the hand: inspection reveals new evidence, binding selects a motor goal, and local control continues during reasoning. Separate image and inspection-policy interventions test the value of new observations and agent-selected views. Our contribution concerns their interaction with continuous in-hand control.

Asynchronous robot policies. Prior work handles inference delay through real-time action-chunk execution [5], online VLA corrections [18], latency-aware RL [22], and temporal counterfactual preferences [23]. LiMA combines sparse semantic decisions with asynchronous dexterous action generation [6]. REVEAL trains small local controllers around a motor policy and an agent with explicit tool actions. We study how observation, target binding and control work together. Asynchronous execution and residual actions alone are not new, and our ablations do not isolate the benefit of explicit controller-goal input.

3. Task and Evaluation

Semantic target. Q1 evaluates six objects in simulation: a marked cube, a wooden die, a duck, a hedgehog, a pig, and a milk carton. Q2 and Q3 use only the marked cube. Instructions request one of six surface letters, or one through six dots on the die. The cube letters are $\{T, E, R, L, S, H\}$. Each target has a fixed reference orientation $Q(g)$ that places

the requested face upward. Scoring compares the full quaternion, including rotation about the face normal. This rule is shared by all methods.

At step t, the agent receives an RGB image I_t and instruction ℓ. The motor policy receives proprioception x_t and measured pose $\hat{p}_t$. Cube pose is estimated from RGB; other objects use noisy measurements constructed from simulator pose. Controllers do not directly receive the unperturbed true pose p_t. Teacher actions and privileged targets are used for training, and true pose is used for independent scoring. During deployment, the controller cannot use a future agent reply. We denote the active goal vector by g_t, the recurrent state by h_t, and a pending request by $m_t = 1$.

Timing. Only one agent request can be pending at a time. Cube episodes allow up to six model calls and 3,600 control steps at 30 Hz. Other objects use 20 Hz over the same 120 seconds of simulated time. For the cube, a call taking L wall-clock seconds can affect control only after it completes and at least $\lceil 30L \rceil$ steps have passed since the request. Physics continues while the agent reasons. The simulator runs slower than real time, so this models inference delay but does not demonstrate a 30 Hz hardware system.

Autonomous completion. All cube systems use the same stopping check at 3 Hz. The agent must have bound the requested symbol to a goal, with no request pending and no inspection underway. Measured orientation error must be at most $\epsilon = 0.16$ radians. This check can stop the system without an explicit stop action from the agent. On other objects, REVEAL uses calibrated completion correction, while the baselines use a measured-pose check. Their stopping rules are detailed in the supplement, and all are scored with the same true-pose tolerance. At stop time τ, success is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared target binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. The target must be correct at the stop time, not merely reached earlier. We do not require an additional hold after stopping. A wrong binding or an incorrect final pose counts as failure even if the object remains in the hand. A drop terminates the trial when the object is at least 0.24 metres from its nominal grasp position.

Paired trials and service failures. Unless stated otherwise, each condition uses 40 trials: 20 per seed under two seeds. Compared methods share the same initial physical states and scored targets. Control and feedback experiments also share instructions. In the mapping experiment, the requested letter changes with the layout so that the physical target stays the same. A service error or a model call taking more than 120 seconds excludes that trial from the service-eligible comparison. We remove the same trial from every compared method, retain all raw results, and do not replace excluded trials. Each reported rate uses the denominator for its own comparison. Service failures are reported separately from control failures.

4. Method

REVEAL builds on a teacher–student motor system adapted from ViserDex [4] for the Revo3 hand. We use its pretrained teacher and student as the motor foundation, adapt the student to a shared goal bank under frozen-teacher supervision, and then train local controllers around the adapted student. At deployment, the agent selects inspection and presentation goals from new observations. Figure 2 separates privileged supervision, student adaptation and closed-loop execution. We describe the cube system below; other objects use their own students, noisy pose feedback and maintenance controllers.

4.1. Visual pose estimation and motor policy

A frozen CNN takes a 120×120 RGB crop and predicts image coordinates and depth for the object center and eight keypoints. Camera calibration and rigid fitting recover object position, XYZW orientation and a fitting error. The rendered cube images include hand occlusion. The deployed policy receives this estimated pose, without ground-truth pose.

We start from a pretrained student that takes geometric goals. A learned projection replaces these goal inputs with $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains proprioception and history without the target pose, and $u_t \in \mathbb{R}^{21}$ contains proposed joint targets in normalized coordinates. We reuse compatible encoders, recurrent and actor weights, and normalization statistics from the released policy. The adapted student receives neither the target quaternion nor the relative-goal quaternion.

We train the goal projection and adapt the recurrent policy using actions from the frozen teacher in a DAgger-style procedure [17]. During adaptation, a training-only decoder reconstructs normalized privileged state targets, with the goal quaternion removed. Its MSE has weight 0.2 alongside the action MSE; goal-projection warmup uses action supervision alone. The decoder and teacher are not used at deployment. The cube student then uses the frozen RGB pose estimator. Object-specific interfaces are described in the supplement.

4.2. From observations to motor goals

The cube has a bank of six normalized goal vectors $B = \{g_1, \ldots, g_6\}$, computed from cached SigLIP2 text fea-

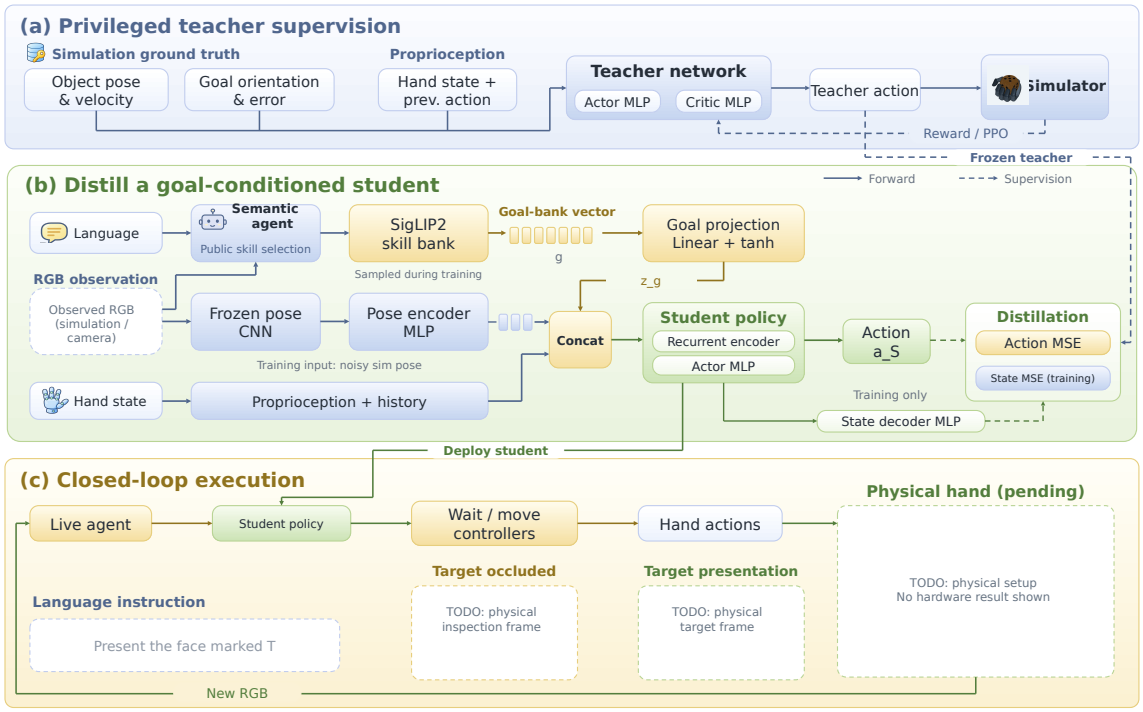


Figure 2. **REVEAL training and deployment.** (a) The motor foundation follows ViserDex's teacher–student approach, adapted to Revo3. (b) A frozen teacher supervises goal-bank-conditioned student adaptation; the state decoder is used only in training. (c) The agent selects inspection or presentation skills, with local controllers adjusting joint commands. Dashed photo slots await physical-hand results.

tures [19]. Each vector represents a color skill and its reference object orientation.

Semantic agent interface. Given the instruction and current image, the hosted agent returns an explicit action: inspect, reorient, verify, stop, or abstain. We validate the action before use. Its color argument selects a goal vector from B. An inspection can move the object toward a new view without declaring that the requested target has been found. The agent chooses inspection and reorientation durations of 10–90 steps, separately from inference latency. It uses new images and the recorded action history for later decisions. Tool arguments select the student goal through a fixed lookup, without hidden agent features.

4.3. Control while the agent is reasoning

The hand needs updated commands while an agent request is pending. The waiting controller receives a feature vector s_t containing proprioception, measured pose, the student's recurrent state and proposed action, fitting error, binding status and request status. It also receives the pose and command recorded when the request began, which we call the *request anchor*, and the elapsed waiting time. The active goal g_t is encoded separately. Before target binding, this goal represents the startup or inspection skill. The controller

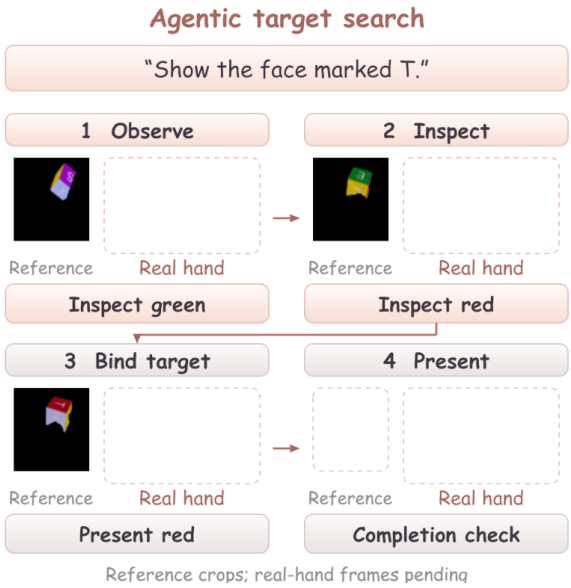


Figure 3. **Visual target search.** The agent observes, inspects candidate faces, binds the requested target, and issues a presentation command. The reference crops are simulated; real-hand frames are pending.

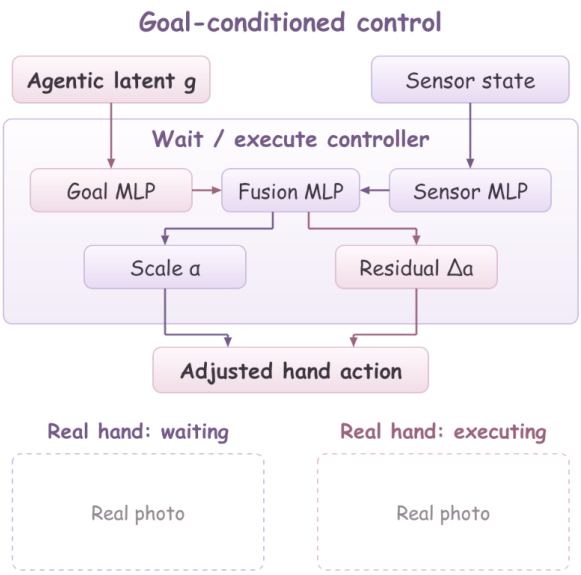


Figure 4. **Local control.** The controller combines sensor and goal features to adjust the speed and direction of the student's joint command. Empty photo slots are reserved for physical-hand results.

does not know the target that a future reply may select.

A fusion layer combines the outputs of a sensor encoder and a goal encoder. The waiting actor predicts a speed logit r_t and correction values $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ denote normalized current joint positions and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{4}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{5}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{6}$$

Zero actor outputs give fixed slowing with coefficient 0.15 while a request is pending. The environment keeps the same joint mapping and smoothing. Unlike scalar slowing, the joint correction can change the command direction as well as its size.

During training, the frozen teacher provides holding actions for the request anchor. At deployment, the actor uses only measured inputs and the active goal, without the pending reply. Writing the correction as $\delta_\phi = 0.35\tanh(v_t)$, we train it to match the teacher action while keeping this correction small:

$$\mathcal{L}_{\mathrm{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{7}$$

The waiting-actor update changes only control during pending requests. The student, pose estimator, execution controller and stopping rule remain fixed. Training details are in the supplement.

4.4. Learning execution after a decision

When no request is pending ($m_t = 0$), a separate execution actor supplies v_t in Eq. 6. It learns from teacher actions to add a bounded correction to the student's proposal. The actor also receives the active goal, although our experiments do not isolate the benefit of this extra goal input.

4.5. Closed-loop execution

At each control step, REVEAL updates the pose estimate, runs the recurrent student, selects the waiting or execution actor, and sends the resulting joint targets to the hand. Only a valid agent reply can change the active skill or target binding. The next image shows the object after the motion that occurred during inference. A separate stopping check determines whether to end the trial. Invalid replies cannot select the true target through an oracle fallback.

We evaluate student adaptation separately from controller learning because a new student also changes the recurrent features and action proposals seen by the controllers.

5. Experiments

We first examine active target search (Q3), then control during reasoning (Q2), and finally complete-system performance across objects and instructions (Q1). Q2 and Q3 use the marked cube. Q4 describes the pending hardware evaluation.

5.1. Setup and reporting

We use Revo3 assets in Isaac Lab 3.0/Newton. Unless specified otherwise, each condition has 40 trials across two evaluation seeds, paired by physical task. Cube systems share the frozen RGB pose estimator, goal bank, student, stopping rule and 0.16-radian tolerance. Other objects use separate students and noisy simulator pose. REVEAL additionally uses calibrated completion correction on these objects. The VLM baselines omit the extra control actors and follow a fixed inspection schedule when the target is unreadable. REVEAL chooses its inspection actions. These differences and measured inference delays are part of the complete-system comparison.

The current cube system updates only the waiting actor. Q1 and Q3 compare methods on the same service-eligible trials. Q2 reports all 120 trials per method. Q1 and Q2 use different trial sets and answer different questions, so their absolute success rates need not match. Controller gains are measured within Q2. The supplement gives the paired tests and corrections for multiple comparisons used in each study.

Condition	Success	%	Drops	Calls
(a) Continued feedback: one vs. up to six calls				
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120
(b) Fresh visual evidence: same six-call cap				
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98
(c) Inspection policy: same model and controllers				
Fixed sweep	31/40	77.5	1	126
Adaptive inspection	38/40	95.0	0	117

Table 1. **Q3: active target search on the cube.** Each panel has 40 scheduled trials per condition. (a) excludes one service-affected pair and reuses Q2's initial full-feedback subset. (b) changes only the agent's RGB input. (c) changes inspection direction and duration. (b,c) have no exclusions. Drops/calls include all trials; actual inference cost differs.

5.2. Q3: Active Target Search and Visual Feedback

Three separate paired experiments examine the search process (Table 1): continuing to observe and decide, receiving new images, and choosing where and how long to inspect. Each comparison uses its own trial set.

(a) Does continued feedback help find the target? The initial plan selects up to six bounded actions from one image and measured feedback. Full feedback allows up to six calls with new images and history. Both use the same local controllers, stopping rule and time limit. Success rises from 6/39 to 33/39 eligible trials: 27 pairs succeed only with feedback and none only with the initial plan (exact $p = 1.49 \times 10^{-8}$). One service failure excludes a matched pair. Of 28 initial-plan abstentions, 22 succeed with feedback; 32 feedback trials first bind after inspection. Figure 5 illustrates this search process. This comparison also changes the call budget.

(b) Does the next image provide useful evidence? Both conditions allow six calls with the same request rule, live numeric feedback, history and local control. Only the agent's RGB input changes: the current image versus the exact first frame on every call. Success is 32/40 versus 9/40, with all 23 pairs with different outcomes favoring fresh images (exact $p = 2.38 \times 10^{-7}$). Correct terminal bindings number 34 versus nine. Both conditions first bind correctly on call one in the same nine trials. Fresh images enable 25 additional correct bindings later, of which 23 lead to success; repeated first frames enable none. Drops differ by only one. These records connect new observations to target binding and completion. They measure declared bindings, not initial visibility or first visual recognition.

Waiting policy	Success	%	Wait drops	Exec. drops	Other
Continue student	73/120	60.8	24	14	9
Fixed slowing	88/120	73.3	13	8	11
Learned slowing	82/120	68.3	7	14	17
Slowing + correction	**104/120**	**86.7**	**5**	**4**	**7**

Table 2. **Q2: controller ablation on the cube.** Each method has 120 trials: 40 per evaluation seed, with trained weights fixed. Wait and execution drops denote failure stages; Other contains the remaining failures. Outcome counts sum to 120 in each row.

(c) Does choosing the next view help? Forty new pairs compare agent-selected inspection directions and durations with a fixed color sweep of 90-step actions. Both use the same model, controllers, fresh images, history format and stopping rule. Feedback reports the action actually executed. The model still decides whether to inspect, bind or abstain in both conditions. Adaptive inspection achieves 38/40 successes (95.0%), compared with 31/40 (77.5%) for the fixed sweep. Nine pairs favor adaptive and two favor fixed inspection, giving a 17.5-point observed gain (exact $p = 0.0654$). Correct terminal bindings are 39 versus 32, with zero versus one drop. This suggests a benefit from choosing inspection actions, but the success difference is not significant at $\alpha = 0.05$ and tests only one fixed sweep.

5.3. Q2: Controller ablations on the cube

We compare four ways to control the hand while the agent is reasoning: continue the student's command, slow it by a fixed coefficient of 0.15, learn the slowing coefficient, or learn slowing together with a bounded joint correction. All four use the same execution controller when no request is pending. Each method has 120 trials: 40 under each of three evaluation seeds, with the trained weights fixed. These seeds vary evaluation tasks, not training initialization.

Learned slowing with joint correction achieves 104/120 successes (86.7%), compared with 88/120 (73.3%) for fixed slowing, a gain of 13.3 percentage points. Waiting drops decrease from 13 to five, and execution drops from eight to four. Learning speed alone gives 82/120 successes (68.3%), below fixed slowing. The results favor learning speed and joint correction together. Drop stages tell us when failure occurs, but do not identify its cause. Seed settings and outcome breakdowns are in the supplement.

5.4. Q1: Systems across objects and mappings

We compare original and changed target-to-skill mappings on six objects (Table 3). Earlier native-image studies and their inconclusive REVEAL–fine-tuned comparisons are reported separately in the supplement.

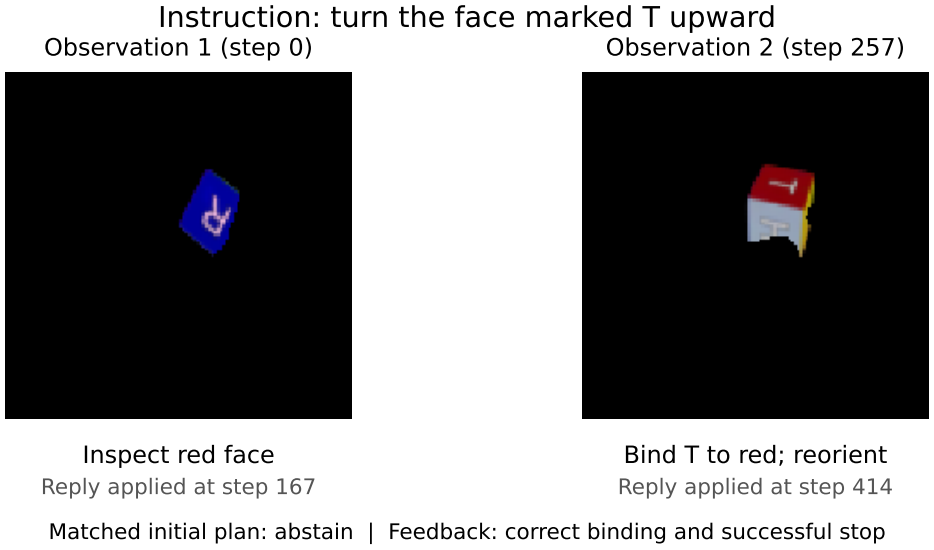


Figure 5. **Recorded inspection before binding.** Unretouched model-input frames from a recorded simulated trial. The initial plan abstains; feedback reveals T, then REVEAL binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from identical state and image. This selected example illustrates evidence acquisition, not wrong-binding correction or hardware performance.

Object	Motor pose	$n_{\text{orig}}/n_{\text{changed}}$	Original mapping			Changed mapping		
			Rule	Fine-tuned	REVEAL	Rule	Fine-tuned	REVEAL
Cube	RGB pose	40 / 40	57.5	75.0	**80.0**	57.5	40.0	**80.0**
Die	Noisy pose	40 / 40	32.5	57.5	**70.0**	2.5	0.0	**65.0**
Duck	Noisy pose	38 / 40	39.5	36.8	**65.8**	22.5	5.0	**60.0**
Hedgehog	Noisy pose	39 / 40	43.6	33.3	**71.8**	25.0	5.0	**80.0**
Pig	Noisy pose	40 / 40	52.5	40.0	**92.5**	37.5	7.5	**82.5**
Milk carton	Noisy pose	40 / 40	32.5	22.5	**42.5**	7.5	2.5	**27.5**
Mean	Object-specific	–	43.0	44.2	**70.4**	25.4	10.0	**65.8**

Table 3. **Q1: six objects and two semantic mappings.** Completion (%) uses common service-eligible trials; each cell has 40 scheduled trials. Mean weights six objects equally. Bold denotes the highest observed rate, not significance. Object-specific weights remain fixed across mappings; bare baselines omit added control modules. Cube uses reconstructed upper images and RGB pose; die rotates its visual mesh; other objects permute surface letters with noisy simulator pose. Native-image and language cohorts remain separate.

Aggregate system performance. Across 1,440 scheduled trials, mean success is 70.4% with original mappings and 65.8% with changed mappings. The strongest baseline averages are 44.2% for fine-tuning and 25.4% for rules, respectively. REVEAL improves on them by 26.2 and 40.4 percentage points. These post-hoc averages weight each object's service-eligible rate equally. They do not pool trials or provide an additional significance test.

Four additional objects. Duck, hedgehog, pig and milk carton contribute 960 trials across two mappings. REVEAL has the highest observed rate in each condition (Table 3), with a separate policy for each object. Three service failures exclude matching trials from all methods. The milk carton still drops in 15/40 original-mapping and 20/40 changed-mapping REVEAL trials. Thus, higher completion does not guarantee a stable grasp. Reset checks and sensitivity analyses are in the supplement.

Visual-semantic intervention. We change only the agent's RGB, keeping pose-estimator input and lighting fixed. Each method uses 40 matched states with original, reassigned or hidden letters, all reconstructed consistently. Reassignment leaves REVEAL at 32/40 successes but reduces fine-tuning from 30/40 to 16/40 (Holm $p = 0.0130$, six comparisons); rules remain at 23/40. Hiding letters gives 0/39, 3/40 and 0/40, respectively. Nine of eleven fine-tuned wrong bindings follow the old mapping, suggesting reliance on learned associations. One reassignment does not establish broad generalization.

System	English		Chinese	
	Standard	Rewrite	Standard	Rewrite
Cube (n)	56	59	57	53
Rule VLM	48.2	45.8	31.6	32.1
Fine-tuned VLM	71.4	72.9	75.4	**81.1**
REVEAL	**75.0**	**79.7**	**84.2**	67.9
Duck (n)	60	59	60	60
Rule VLM	50.0	25.4	33.3	31.7
Fine-tuned VLM	38.3	30.5	31.7	35.0
REVEAL	**68.3**	**67.8**	**68.3**	**76.7**
Hedgehog (n)	60	60	52	60
Rule VLM	46.7	38.3	36.5	36.7
Fine-tuned VLM	38.3	31.7	34.6	25.0
REVEAL	**73.3**	**76.7**	**75.0**	**76.7**

Table 4. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed n of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

Dice layout transfer. The 240-trial die study rotates the visual mesh while preserving collision geometry, inertia and the skill-ID prompt. Under the changed layout, exploratory paired tests favor REVEAL over fine-tuning ($p = 2.98 \times 10^{-8}$) and rules ($p = 5.96 \times 10^{-8}$). REVEAL's own layout difference is inconclusive ($p = 0.688$). These are complete-system comparisons with different inspection, history and control modules; the supplement reports all seven unadjusted contrasts.

Language and wording coverage. We test standard and rewritten instructions in English and Chinese on three objects: 60 trials per condition across three seeds, totaling 2,160 trials. Table 4 compares the same service-eligible trials for all methods. We keep weights, mappings and scoring fixed, without retraining for Chinese. The rule baseline also translates its task/output instructions. This tests language coverage of the deployed systems. The supplement retains all raw outcomes, including service failures.

REVEAL reaches 67.8–76.7% on duck and 73.3–76.7% on hedgehog, exceeding fine-tuning in all eight conditions (Holm-adjusted paired $p \leq 0.0258$ over 36 method tests). Six of eight differences against rules also survive correction. Cube comparisons remain inconclusive: fine-tuning leads for rewritten Chinese (81.1% versus 67.9%), while REVEAL leads in the other three conditions, all adjusted $p = 1.0$. Strict state/image pairing preserves these significance decisions. Language/wording tests are inconclusive after a separate 36-test correction.

5.5. Q4: Multi-object hardware evaluation

Hardware protocol. **[TODO: Complete the multi-object physical-hand evaluation.]** No hardware outcomes are available yet. The planned comparison uses REVEAL and two baselines per object, reporting completion, drops, false stops and inference cost. The supplement details calibration, resets, independent scoring and synchronized videos.

6. Discussion and Limitations

Search and control. The strongest mechanism evidence concerns target acquisition: fresh images enable correct bindings that are absent with a repeated first view. Choosing inspection actions shows a favorable but statistically unresolved trend. Q2 separately supports joint correction while the agent reasons. Together, these studies support the search–binding–control loop. Q1 extends the evaluation to six object-specific systems, but its differences also include model, history, controller and stopping choices.

Evaluation limits. Equal call limits do not imply equal inference cost. Controller-goal conditioning has not been isolated, and Q3 tests one fixed sweep on one cube. Success measures correct binding and pose at termination. Independent target visibility and sustained presentation after stopping remain unmeasured; these are distinct from the reported completion rate.

Deployment scope. Each object uses separate weights. Cube pose comes from rendered RGB; other objects use noisy simulator pose. Real-camera performance, transfer to new objects and hardware execution remain untested. Hosted models may change over time, limiting reproducibility. Rates excluding service failures do not measure end-to-end availability.

7. Conclusion

REVEAL connects active target search and semantic binding to continuous in-hand control, using a ViserDex-inspired teacher–student foundation adapted to Revo3. Cube experiments show that new observations enable additional target bindings and that joint correction supports control during inference. Adaptive inspection yields a promising but statistically unresolved gain over a fixed sweep. Across six object-specific systems, mean success is 70.4% with original mappings and 65.8% with changed mappings. Physical-hand tests and direct visibility and hold measurements remain necessary to assess sustained target presentation.

LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. A hosted vision-language model and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

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